

Received 8 November 2024, accepted 24 December 2024, date of publication 27 December 2024, date of current version 28 January 2025.

Digital Object Identifier 10.1109/ACCESS.2024.3523491

APPLIED RESEARCH

A Novel Approach to Cognitive Load Measurement in N-Back Tasks Using Wearable Sensors, Empirical Mode Decomposition With Machine Learning, and Explainable AI for Feature Importance

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This work was supported by the Ministry of Science and ICT (MSIT), South Korea, under the National Program for Excellence in Software, Supervised by the Institute of Information and Communications Technology Planning and Evaluation (IITP) in 2022 under Grant 2022-0-01091 and Grant 1711175863.

ABSTRACT This study introduces a novel approach for detecting mental workload and stress, utilizing ECG and Fingertip-PPG data from the MAUS dataset. The dataset includes physiological recordings such as ECG, Fingertip-PPG, Wrist-PPG, and GSR signals from 22 participants exposed to varying levels of mental workload and stress conditions, serving as a valuable resource for developing mental workload assessment systems. Focusing on ECG and Fingertip-PPG signals, we evaluated their effectiveness in detecting mental workload and stress. Early detection of mental workload and stress is vital for preventing adverse health outcomes, emphasizing the importance of timely intervention. In this study, by applying Empirical Mode Decomposition (EMD) for feature extraction and machine learning classifiers, we achieved high performance with accuracy rates of 88.64% for ECG and 80.68% for Fingertip-PPG Signal. Combining both data types (ECG and Fingertip-PPG data) further improved recall to 94.87% and boosted overall accuracy to 87.50%, marking an improvement of approximately 13%. SHAP analysis identified key features contributing to this performance, revealing key features such as the waveform length of the first Intrinsic Mode Function (IMF) for the ECG signal, the mean of the second IMF for fingertip-PPG, and the waveform length of the second IMF for combined ECG and fingertip-PPG signals. These results highlight the potential of combining physiological signals and EMD with machine learning for detecting mental workload and stress.

INDEX TERMS Physiological signals, electrocardiogram (ECG) signal, photoplethysmography (PPG) signal, machine learning, empirical mode decomposition, mental workload, stress detection, Shapley additive explanations, intrinsic mode function.

I. INTRODUCTION

In contemporary society, stress has evolved into a ubiquitous issue affecting individuals across various domains of life, requiring precise identification and practical solutions.

The associate editor coordinating the review of this manuscript and approving it for publication was Lorenzo Mucchi¹.

Recent technological advancements and data analytics have opened avenues for innovative approaches to mental workload and stress detection, with a notable focus on utilizing physiological signals such as Electrocardiogram (ECG) and Photoplethysmography (PPG).

Furthermore, monitoring mental workload (MW) has emerged as a critical necessity in numerous contexts,

including safety measures and smart technology applications like driver awareness and Brain-Computer Interfacing (BCI). MW assessment offers insights into operator processing capabilities and subjective psychological experiences [1], crucial for minimizing errors and preventing “overload” conditions. Physiological signals, including ECG, PPG, and Galvanic Skin Response (GSR), serve as valuable indicators for MW assessment [2], [3].

Most prior studies have focused on identifying mental workload and stress via physiological signs. Several studies have used multiple physiological signals such as electrocardiogram (ECG), electroencephalogram (EEG), galvanic skin response (GSR), electromyogram (EMG), and arterial blood pressure (ABP) to identify stress, either in binary form (stress/no stress) or across multiple levels (e.g., low, moderate, and high) [4], [5], [6]. These studies contend that a single marker cannot fully quantify an individual’s mental workload and stress response [7]. In contrast, several previous studies used a single signal, often ECG or GSR, to identify stress levels [8], [9], [10], noting benefits such as cheaper computing costs, enhanced real-time system performance, and the utilization of conventional wearable device data [11]. Generally, ECG or Heart Rate Variability (HRV) signals are useful stress markers.

The low cost of sensors has permitted the development of commercial wearable devices that monitor and record a variety of physiological signals, including photoplethysmography (PPG), galvanic skin response (GSR), and electrocardiography (ECG). These signals are easily captured in basic configurations using only a few electrodes. PPG is preferred for assessing mental workload and stress and tiredness levels with a single sensor mounted on a finger or wrist. PPG signals also allow blood pressure, heart rate, and saturation levels to be monitored. These advantages make PPG useful in a wide range of applications and goods. This explains the enormous research interest in PPG and the numerous studies that use PPG signals to categorize mental stress [12]. These studies use PPG signals alone or in conjunction with other physiological markers. Experiments have shown [13] that specific features of PPG signals might indicate stress levels.

Furthermore, PPG enables the extraction of data on blood pressure [14] and heart rate variability (HRV) [15]. These factors may be extracted from the PPG signal and then utilized to identify elevated stress levels. Furthermore, one unique stress signal is a structural alteration in the shape of PPG pulses generated by psychological stress-induced arterial stiffness.

The MAUS database significantly contributes to meeting the demand for reliable datasets for MW evaluation. This dataset comprises various physiological signals, including single-lead ECG, GSR, fingertip PPG, and wristband PPG, collected under varying MW conditions induced through the Nback task. The N-back task provides a reliable, objective reference for MW assessment, with stimuli intensity correlating to MW levels [16]. In this study, we have employed ECG and fingertip-PPG signals.

Integrating physiological signal processing techniques, such as Empirical Mode Decomposition (EMD) for feature extraction, further enhances the understanding of mental workload and stress responses. EMD facilitates extracting relevant features corresponding to stress-related components, enabling a comprehensive characterization of physiological responses [17]. Furthermore, the fusion of ECG and PPG signals provides enhanced insights into stress detection, allowing a more comprehensive assessment of cardiovascular dynamics and peripheral perfusion [18].

Machine learning classifiers, including Random Forest, Support Vector Machines (SVM), and Gradient Boosting, use annotated datasets to accurately distinguish between stress and non-stress conditions, allowing real-time monitoring and intervention [19].

To improve the interpretability of mental workload and stress detection algorithms, Shapley Additive explanations (SHAP) analysis provides insights into the contribution of individual features, enabling actionable insights from underlying physiological data [20].

Our study utilized ECG and fingertip-PPG signals due to their established role as reliable physiological mental workload and stress indicators. ECG, in particular, provides insights into cardiac activity and autonomic nervous system response, whereas PPG captures variations in blood volume, reflecting vascular dynamics and peripheral circulation. We opted for these signals for their ability to offer a comprehensive understanding of stress-related physiological changes.

Our contributions can be summarized as follows:

1. To provide a robust framework that uses physiological signals to detect stress: ECG and fingertip-PPG signals are used as indicators of physiological arousal and stress. For feature extraction, use Empirical Mode Decomposition (EMD) to identify workload from the signals.
2. To accurately classify stress using machine learning classifiers: Use supervised learning techniques, such as gradient boosting, random forest, and SVM for stress categorization.
3. To improve the stress detection models’ interpretability by the application of SHAP analysis: Utilize Shapley Additive Explanations (SHAP) analysis to identify which feature had the most impact on the model’s predictions.

The overall aim of this project is to provide a comprehensive framework that detects stress by combining machine learning classifiers, physiological signals, EMD feature extraction, and SHAP analysis. This study intends to provide a deeper understanding of mental workload and stress physiology and promote the development of efficient stress monitoring and management solutions by combining advanced signal processing techniques with machine learning methodology.

II. LITERATURE REVIEW

Sethi et al. [21] suggested a supervised learning method for detecting stress using a multimodal MAUS dataset

that included fingertip PPG, GSR, and ECG signals. They attained an F1 score of 73.9% and an accuracy of 75.7% by concatenating normalized ECG and PPG data with the AdaBoost Classifier.

Beh et al. [22] proposed outlier elimination and uncertainty estimate strategies to improve the reliability of PPG-based mental workload assessment systems. Their technique, evaluated using CLAS and MAUS datasets, outperformed ECG-based algorithms regarding accuracy (74.2%) and variance (10.8%).

Feradov et al. [23] studied EMD-based characteristics for stress assessment using PPG data. Using Decision Tree, kNN, and SVM classifiers, they achieved up to 87.8% classification accuracy on the CLAS dataset, highlighting the potential of PPG-based stress detection applications. Radhika et al. [24] studied the combined modality characteristics from EDA and ECG data in the frequency domain to identify stress. Their strategy outperformed previous approaches and demonstrated generalization across four benchmark datasets, proving its efficacy in stress detection.

Kuttala et al. [25] suggested a hierarchical feature fusion strategy based on multimodal EDA, ECG data, and Convolutional Neural Networks (CNN). Their technique, evaluated on four benchmark datasets, outperformed previous models, notably on frequency band characteristics.

In [26], Heo et al. used a two-step de-noising approach paired with ensemble-based multiple peak-detecting techniques for feature extraction. They tested these characteristics in an experimental context using a variety of classification algorithms for both three-class and quad-class classification tasks, attaining an accuracy of up to 96.5%.

Karla and Sharma [27] studied using neural networks to categorize mental stress levels in an experimental setting with five classes: relaxation, deep breathing, and three levels of mental tension. They investigated the application of 18 characteristics, 9 each from the time and frequency domains, and obtained a classification accuracy of 91.1%.

Another study [28] used heart rate variability and Discrete Wavelet Transform (DWT) coefficients to create a real-time stress assessment system. Using an AdaBoost classifier, they attained a classification accuracy of 93%.

Zhuang et al. [29] provide a technique for emotion identification based on empirical mode decomposition (EMD) of EEG data. They extract multidimensional characteristics from Intrinsic Mode Functions (IMFs) to accurately identify emotions. The paper examines the role of each IMF and compares the suggested method to traditional methodologies, exhibiting better performance on DEAP datasets.

Samal and Hashmi [30] use EEG data to examine ensemble median empirical mode decomposition (MEEMD) for emotion identification. They propose a MEEMD-based feature extraction approach that achieves 74.3% valence and 78% arousal accuracy on the DEAP EEG emotion dataset.

Zhang et al. [31] offer a method for categorizing emotions from EEG data that employs empirical mode decomposition

(EMD) and an autoregressive model. By extracting features from IMFs and using an AR model, they obtain an average recognition rate of 86.28% for four binary-class tasks on the DEAP dataset, outperforming existing techniques.

Mert and Akan [32] explored the application of multi-variate empirical mode decomposition (MEMD) in emotion identification using EEG data. Due to the dynamic nature of EEG signals, advanced signal processing techniques are required. MEMD, a variant of EMD, breaks down multichannel EEG data into intrinsic mode functions (IMFs). Several temporal and frequency domain methodologies are utilized to examine the Intrinsic Mode Functions (IMFs) as characteristics for measuring valence and arousal levels. The suggested technique is assessed using the DEAP emotional EEG dataset and compared to earlier research for benchmarking purposes.

III. MAUS DATASET

The MAUS dataset was methodically developed to compile a collection of physiological signals to investigate a mental workload (MW) evaluation methodology designed specifically for wearables. MW, defined as the proportion of an operator's cognitive resources required to fulfill system demands, emerges as a critical concept in understanding human-computer interaction dynamics [33]. The MAUS dataset, which includes ECG, Fingertip-PPG, Wrist-PPG, and GSR signals, is valuable for studying MW responses across many physiological modalities.

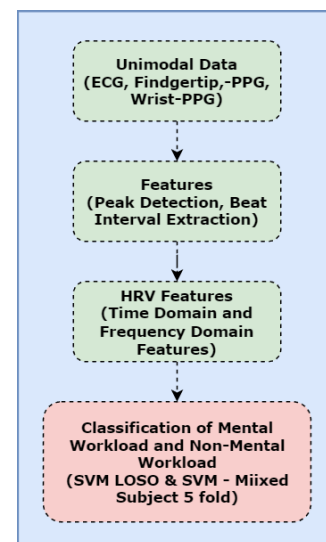


FIGURE 1. Pipeline of mental workload classification.

A. MODALITIES AND STIMULI

The collection contains recordings of ECG, wrist-PPG, fingertip-PPG, and GSR physiological signals. To further enhance the depth of MW evaluation, it incorporates subjective indicators such as the NASA Task Load Index

(NTLI) and the Pittsburgh Sleep Quality Index (PSQI). The dataset, which includes 22 participants (20 males and two females), records various physiological reactions to cognitive stimulation. The N-Back Test is a crucial component of the experimental paradigm. It challenges participants to memorize a sequence of rapidly flashing numbers and recall the last N digits when they recognize a stimulus that corresponds to the n th number before it. This cognitive activity serves as a controlled stressor, eliciting varying degrees of MW, and increases in complexity with each iteration ($0 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow 0$). Participants complete the PSQI and NTLI questionnaires before and after the task, providing valuable insights into their subjective MW appraisal and sleep quality, respectively.

B. FEATURE EXTRACTION

The methods for feature extraction involve meticulous peak detection techniques customized for every physiological input. PPG signals are filtered with a median value of 13 before peak identification, whereas ECG signals are filtered using a bandpass range of 3 to 45 Hz and then R-peak detection. The characteristics that were extracted include time-domain metrics (like SDNN, NN50, and RMSSD) as well as frequency-domain metrics (like Total Frequency, High Frequency, and Low Frequency) that provide a thorough picture of the body's reactions to microwave radiation.

C. CLASSIFICATION

Fig. 1 shows the process for classifying machine learning in an unimodal method, explaining the approach taken for feature extraction and subsequent classification. SVM forms the basis of binary MW classification based on frequency domain and HRV statistical variables taken from the MAUS dataset. The evaluation of classification accuracy is supported by strong validation methods such as mixed subject 5-fold cross-validation and Leave-One-Subject-Out (LOSO). Additionally, average accuracies for ECG, wrist PPG, and fingertip PPG modalities are 71.6%, 66.7%, and 59.9% when using SVM with LOSO validation [16]. Performance measures that clarify the effectiveness of the classification models include confusion matrix, accuracy, F1 score, precision, and recall [16], [21].

IV. TECHNIQUES FOR STRESS DETECTION

A. EMPIRICAL MODE DECOMPOSITION

Empirical mode decomposition (EMD) is a signal processing technique for feature extraction in physiological signals. It decomposes a signal into what are known as intrinsic mode functions (IMFs), which are a limited collection of oscillatory components. Each IMF is a unique representation of the signal's local dynamics, reflecting a specific oscillatory mode inherent in it. Mathematically, the EMD process can be expressed as follows: Given a signal $x(n)$, the decomposition process involves extracting a set of IMFs $c_i(n)$ and a residue

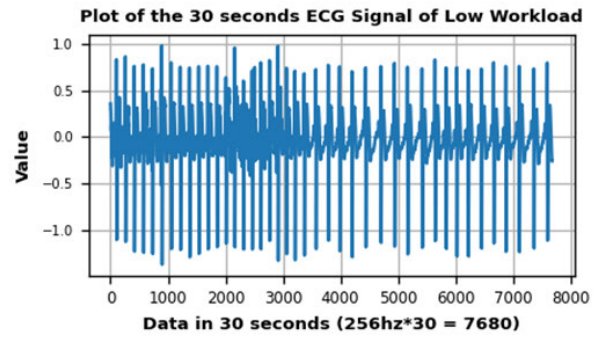


FIGURE 2. Plot of the 30 seconds ECG signal of low workload.

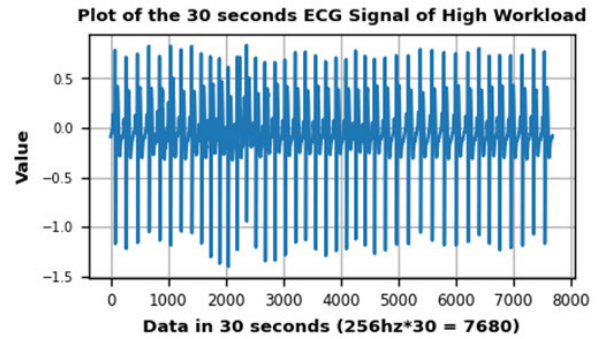


FIGURE 3. Plot of the 30 seconds ECG signal of high workload.

$r(n)$ such that:

$$x(n) = \sum_{i=1}^N c_i(n) + r(n). \quad (1)$$

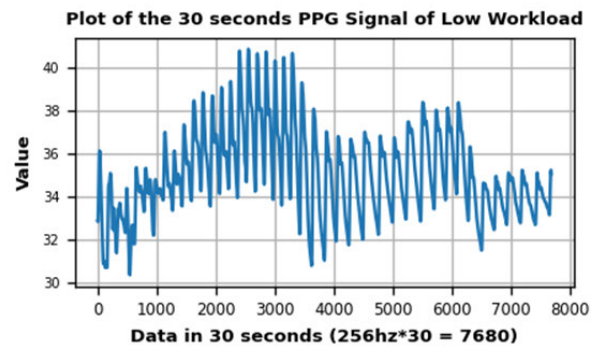


FIGURE 4. Plot of the 30 seconds Fingertip-PPG signal of low workload.

$c_i(n)$ represents the i -th Intrinsic Mode Function. N is the total number of IMFs obtained. $r(n)$ is the residue, representing the high-frequency noise or trend component of the signal. Each IMF $c_i(n)$ satisfies two conditions: It has the same number of zero crossings and extrema, or at most one difference. The mean value of the envelope defined by local maxima and minima of $c_i(n)$ is zero at any point. The

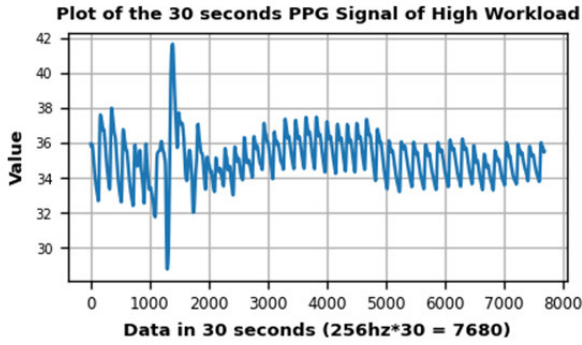


FIGURE 5. Plot of the 30 seconds Fingertip-PPG signal of high workload.

resulting IMFs represent the signal's oscillatory components at different scales, making EMD particularly useful for analyzing signals with complex and non-linear dynamics, such as physiological data. Fig.2-5 shows the Plot of the 30 seconds for ECG and fingertip-PPG Signal of Low and High Workload. Fig.6 shows the Signal and its decomposition into Intrinsic Mode Functions (IMFs) using Empirical Mode Decomposition (EMD) where the first plot shows the original signal with its local maxima and local minima in represented by red and blue respectively including smoothed spline interpolations of these extrema.

B. INTRINSIC MODE FUNCTION

An Intrinsic Mode Function (IMF), a key output of the Empirical Mode Decomposition (EMD) process, is crucial for understanding the local dynamics of a time series signal. Each IMF represents a specific oscillatory mode in the signal, and their symmetric nature around the mean and narrow bands makes them particularly useful for this purpose. The EMD procedure is a step-by-step iterative process that involves the calculation of IMFs. It follows these subsequent steps:

- **Identifying Extrema:** The signal is examined to determine its local maxima and minima.
- **Interpolating Extrema:** Cubic spline interpolation connects the local maxima and minima to create the upper and lower envelopes, respectively.

$$e_{\text{upper}}(n) = \text{CubicSpline}(\text{maxima points}) \quad (2)$$

$$e_{\text{lower}}(n) = \text{CubicSpline}(\text{minima points}) \quad (3)$$

- **Calculating the Mean Envelope:** The upper and lower envelopes are averaged to determine the mean envelope.

$$m_i(n) = \frac{e_{\text{upper}}(n) + e_{\text{lower}}(n)}{2} \quad (4)$$

- **Obtaining the First IMF (IMF1):** The mean envelope is subtracted from the original signal to arrive at the first IMF or IMF1. In this stage, the signal's primary oscillatory mode is isolated.

$$c_1(n) = x(n) - m_1(n) \quad (5)$$

- **Extracting Subsequent IMFs:** To extract the next IMF, the residual signal—the result of subtracting IMF1 from the original signal—repeatedly goes through the same procedure. With each repetition, an extra IMF is produced, capturing oscillations in the signal that are progressively finer-scale. The residual after extracting the $(i - 1)$ th IMF can be defined as:

$$r_{i-1}(n) = x(n) - \sum_{j=1}^{i-1} c_j(n) \quad (6)$$

Then:

- Calculating the mean envelope $m_i(n)$ of $r_{i-1}(n)$.
- Subtracting this calculated mean envelope from $r_{i-1}(n)$ to obtain $c_i(n)$:

$$c_i(n) = r_{i-1}(n) - m_i(n) \quad (7)$$

- Update the residual for the next IMF extraction.

$$r_i(n) = r_{i-1}(n) - c_i(n) \quad (8)$$

The process continues until specific stopping conditions are satisfied, such as completing a predetermined number of iterations or ensuring the residue shows a monotonic trend. Higher-order IMFs capture finer-scale changes, and each extracted IMF reflects a unique oscillatory mode in the original signal. Fig.7-10 shows the Plot of the IMF's for 30 seconds for ECG and fingertip-PPG Signal of Low and High Workload.

V. SHAP FOR EXTRACTED FEATURES

SHAP (SHapley Additive exPlanations) is an XAI (Explainable Artificial Intelligence) technique used to explain the output of machine learning models by attributing the importance of each input feature to the model's predictions. SHAP can be used to analyze the relevance of features retrieved from physiological signals in forecasting mental workload and stress levels or other important outcomes in feature extraction.

Here's how SHAP works:

- **Importance of Each Individual Feature:** SHAP offers a means of measuring how much each feature contributes to the model's output for a certain prediction. This makes it possible to comprehend how every physiological signal component affects the model's choice for a particular data point.
- **Interpretability and Consistency:** SHAP values guarantee that the weights assigned to every characteristic are comprehensible and consistent between various forecasts. This aids in comprehending the relative influence of various attributes on the model's choices.
- **Visualization:** Various charts, including force, dependency, and summary plots, may be used to illustrate SHAP values. These visual aids aid in interpreting each feature's contribution and comprehending the connection between the features and the output of the model.

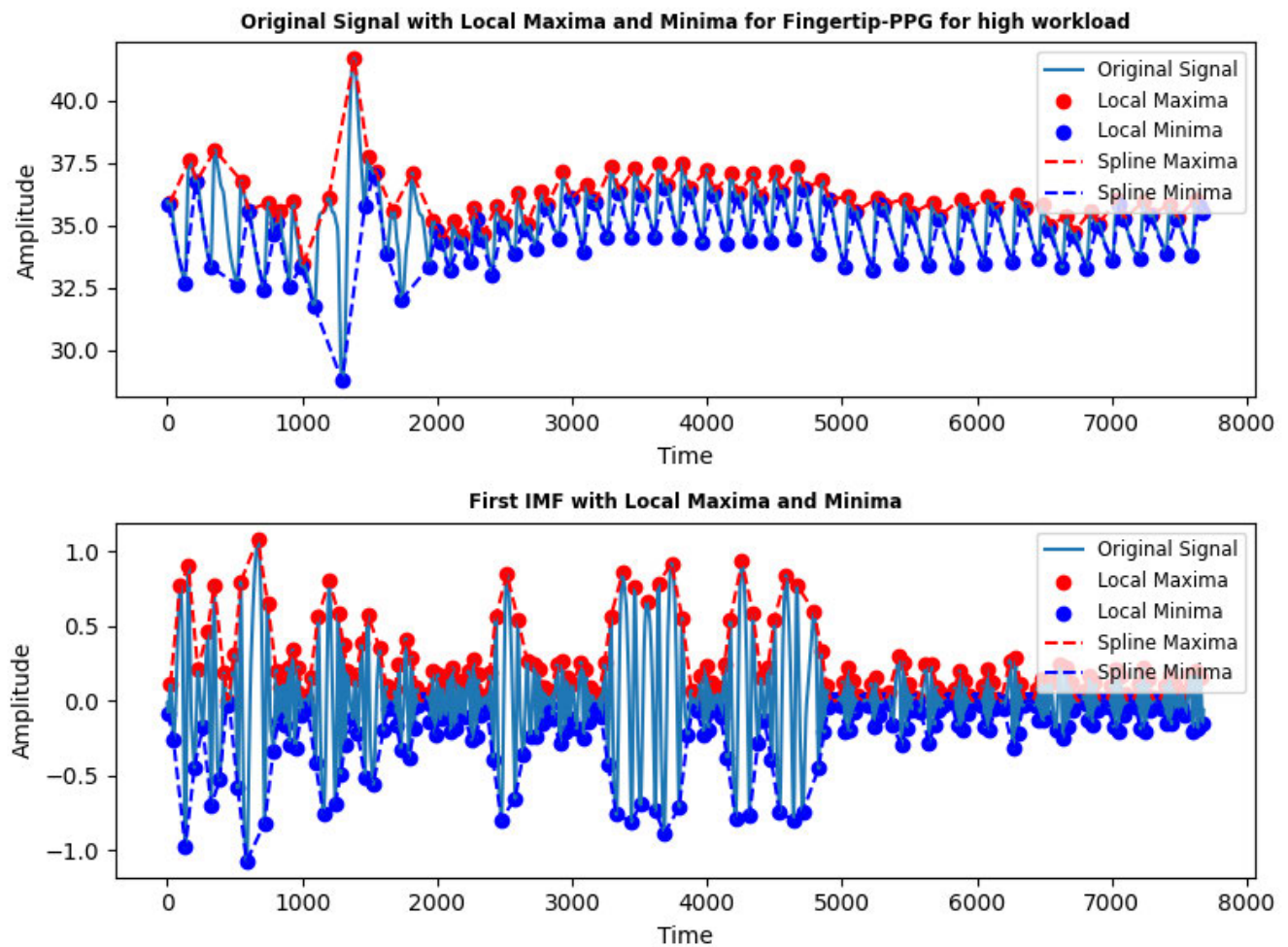


FIGURE 6. 30 second Fingertip-PPG signal and IMF1 with local maxima and minima for high workload.

- **Model Agnostic:** SHAP's ability to be used with any machine learning model, regardless of complexity or type, is one of its main advantages. Because of this, SHAP is a flexible tool for analyzing the forecasts of different models that were trained on data from physiological signals.

When analyzing physiological signal characteristics that have been retrieved, SHAP analysis may be used to determine which aspects are most important for forecasting stress levels or other desired outcomes. For instance, SHAP analysis can identify which particular properties, such as heart rate variability (HRV) or pulse rate, are most important in predicting mental workload and stress levels when data generated from electrocardiogram (ECG) or photoplethysmogram (PPG) signals are used to make these predictions.

In general, SHAP analysis improves the interpretability and reliability of machine learning models used on physiological signal data by offering insightful information about how these models operate internally.

VI. METHODOLOGY

The study was carried out on the MAUS dataset, which aimed at assessing mental workload (MW) through physiological signals, representing a significant step forward in understanding and managing cognitive load in both work and daily life environments. Specifically, the experimental setup used ECG and PPG recordings from the publicly available dataset.

In this study, we have only used the ECG and Fingertip-PPG recording for N-back tasks (Trial 1: 0-back and Trial 5: 3-back).

A. ECG AND FINGERTIP-PPG SIGNAL AS UNIMODAL DATA

We utilized the MAUS dataset, which includes ECG and fingertip-PPG recordings separately, to detect mental workload and stress levels under low and high mental workload conditions using a 30-second window. The data was segmented into 30-second windows, resulting in segments of 7680 samples each, based on a sampling rate of 256 Hz for both fingertip-PPG and ECG. Thirty seconds is sufficient to monitor physiological reactions and changes

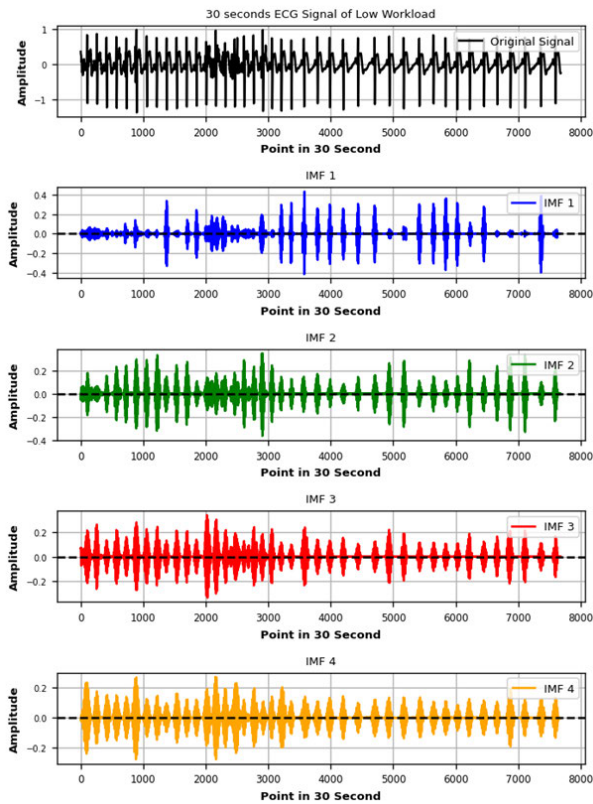


FIGURE 7. IMF's for 30 seconds ECG signal of low workload.

associated with various cognitive states, such as stress levels, without requiring lengthy processing periods. Furthermore, a 30-second frame aligns with the temporal dynamics of physiological processes, ensuring that important patterns and characteristics in the signals are successfully captured.

Each segment of the signal, sampled at 256 Hz (resulting in 7680 samples per segment), is decomposed using Empirical Mode Decomposition (EMD) into four Intrinsic Mode Functions (IMFs). In this study, we applied EMD for feature extraction from the ECG and fingertip-PPG signals, leveraging its ability to handle non-linear and non-stationary data. It decomposes the signal into a finite number of Intrinsic Mode Functions (IMFs), each IMF representing the local oscillatory modes present in the original signal. We specifically selected the first four IMFs as they effectively capture the most relevant physiological information related to mental workload and stress. Testing with more than four IMFs contributed negligible improvements to classification accuracy with the additional computational load. Higher-order IMFs generally included residual noise and trends lacking meaningful physiological changes. Our selection of IMFs aligns with other studies that have used EMD for stress detection [29], [31], where the first few IMFs contained the most significant features.

We focused on the first four IMFs and computed twelve features from each IMF: mean, minimum, maximum, skewness, standard deviation, kurtosis, energy, entropy,

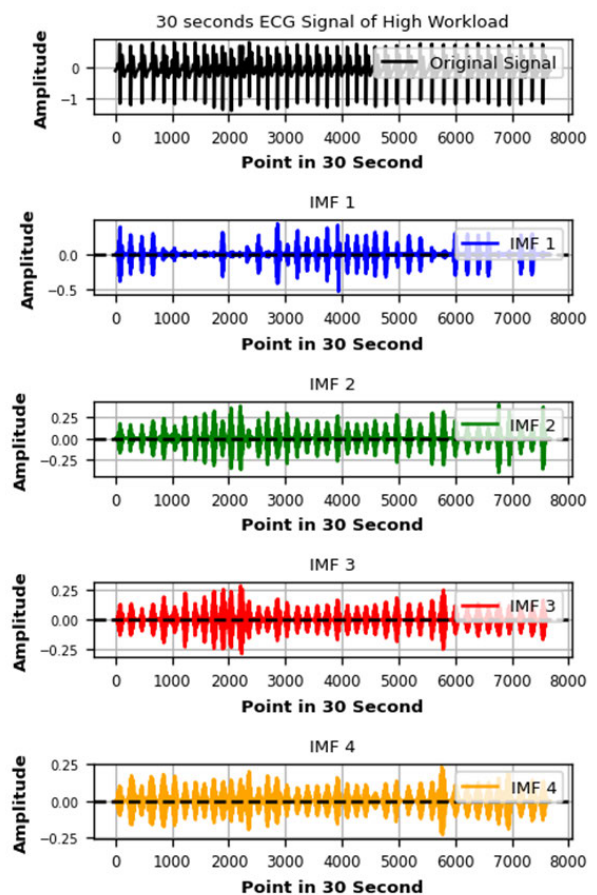


FIGURE 8. IMF's for 30 seconds ECG signal of high workload.

zero crossing rate, peak count, mean absolute value, and waveform length. These features were calculated using functions such as `calculate_entropy` for entropy and `zero_crossing_rate` for the zero crossing rate. The extracted features were labeled as either low (0) or high (1) mental stress based on the experimental conditions. We concatenated the features extracted from low- and high-mental-stress conditions into a single dataset. This involved merging two CSV files containing low and high mental stress features. The final dataset was prepared by dropping unnecessary columns, ensuring a clean and comprehensive feature set for subsequent analysis.

The final dataset was split into training and testing sets with a 70-30 ratio. We evaluated ten different classifiers: Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, AdaBoost, Gradient Boosting, Gaussian Naive Bayes, Linear Discriminant Analysis, Quadratic Discriminant Analysis, and Logistic Regression. Each classifier was trained on the training set and tested on the testing set. We calculated each classifier's accuracy, precision, recall, and confusion matrix. The confusion matrix provided detailed insights into the performance of each classifier.

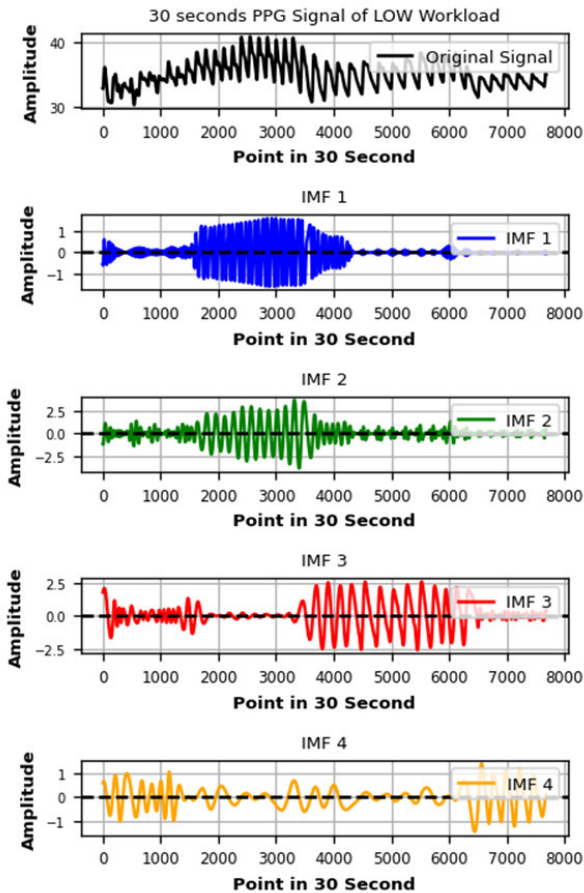


FIGURE 9. IMF's for 30 seconds Fingertip-PPG signal of low workload.

Fig.11 shows the Methodology of ECG Signal as Unimodal Data.

B. ECG + FINGERTIP-PPG SIGNAL AS MULTIMODAL DATA

For the fusion of the ECG + fingertip-PPG signal as shown in Fig. 12, the data was segmented into 30-second windows, resulting in segments of 7680 samples each, based on a sampling rate of 256 Hz for both PPG and ECG separately.

Employing the Empirical Mode Decomposition (EMD) technique separately on both ECG and fingertip-PPG signals yielded a set of Intrinsic Mode Functions (IMFs). We focused on the first four IMFs, based on their ability to capture most relevant information from physiological signals and computed twelve statistical features from each IMF: mean, minimum, maximum, skewness, standard deviation, kurtosis, energy, entropy, zero crossing rate, peak count, mean absolute value, and waveform length. The extracted features were labeled as either low (0) or high (1) mental stress based on the experimental conditions. The final dataset was prepared by concatenating the features extracted from ECG and fingertip-PPG together, dropping unnecessary columns, and ensuring a clean and comprehensive feature set for subsequent analysis.

The final dataset was split into training and testing sets with a 70-30 ratio. Utilizing this concatenated dataset,

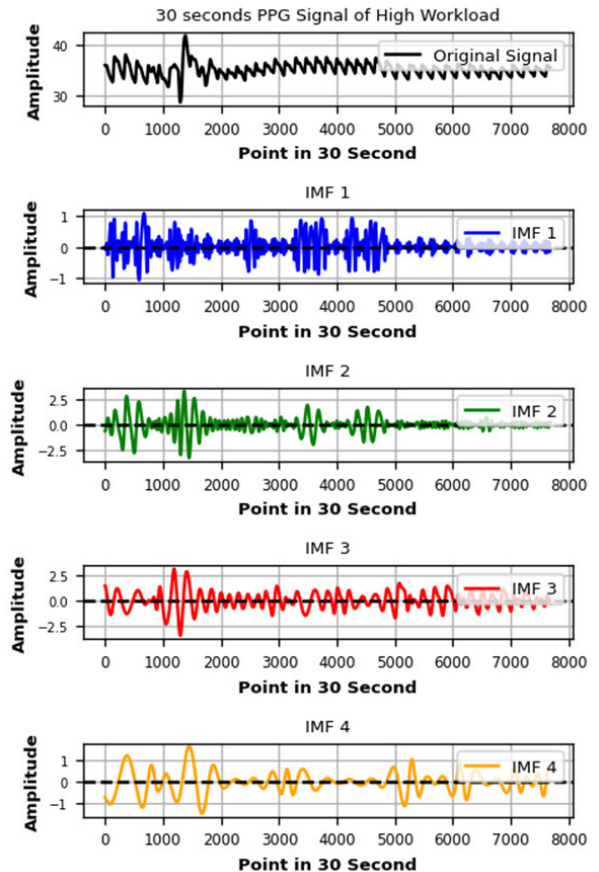


FIGURE 10. IMF's for 30 seconds Fingertip-PPG signal of high workload.

machine learning classifiers including Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, AdaBoost, Gradient Boosting, Gaussian Naive Bayes, Linear Discriminant Analysis, Quadratic Discriminant Analysis, and Logistic Regression were trained and assessed through cross-validation. Evaluation metrics such as accuracy, precision, recall, and confusion matrices were computed.

VII. RESULTS

The performance of various machine learning classifiers was evaluated using accuracy, precision, recall, and confusion matrices. The Random Forest classifier consistently surpassed other classifier models, achieving an accuracy of 88.64% for ECG signals, 80.68% for fingertip-PPG, and 87.50% for the ECG + fingertip-PPG signal as a multimodal dataset. Its ensemble approach allows it to capture complex, non-linear relationships within the data and handle noisy signals more effectively, which outperformed Support Vector Machines (SVM) or K-Nearest Neighbors (KNN). SHAP (Shapley Additive Explanations) analysis identified key features, revealing that the waveform length of the first IMF (Imf_1_WFL) strongly correlated with mental workload for ECG signal. In contrast, the mean of the

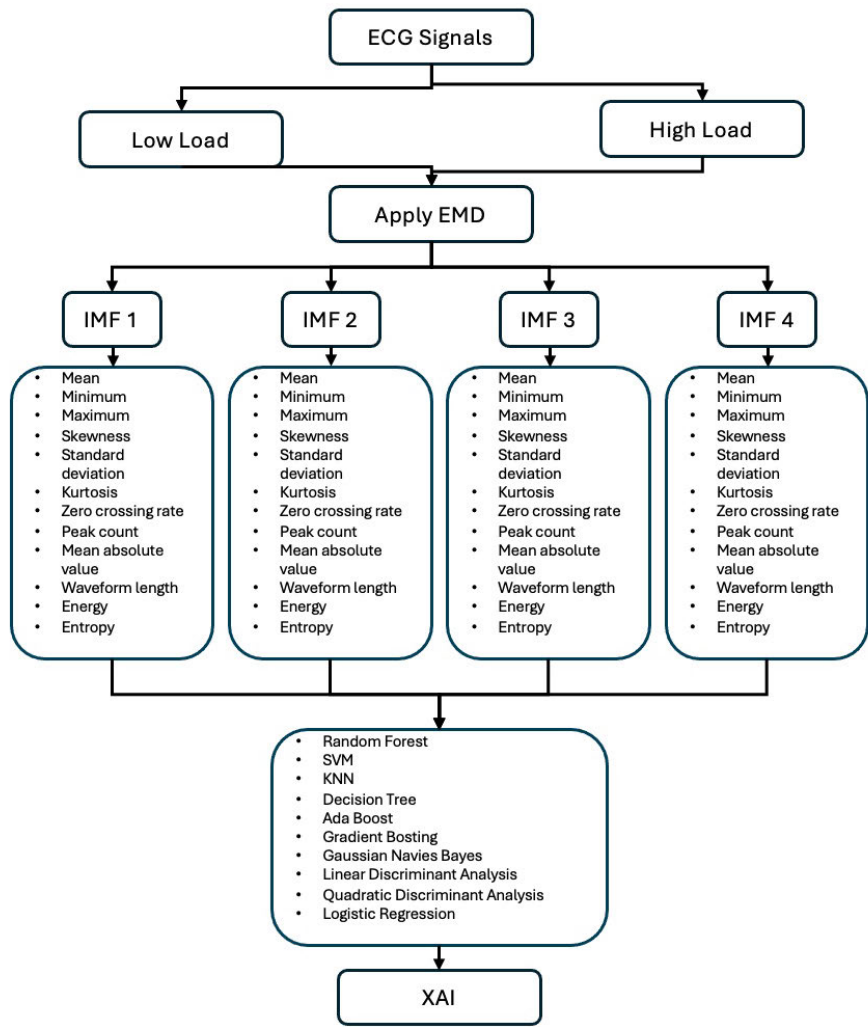


FIGURE 11. Methodology of ECG signal as unimodal data.

second IMF (Imf_2_MEAN) was significant for fingertip-PPG. In the multimodal approach, the waveform length of the second IMF (Imf_2_WFL) emerged as the leading feature, enhancing recall to 94.87%, which increased true positive detection rates.

TABLE 1. Performance metrics of random forest classifier.

	ECG		PPG		ECG+PPG	
Accuracy (%)	88.64		80.68		87.50	
Precision	0.87		0.79		0.80	
Recall	0.91		0.77		0.94	
Confusion Matrix	TP = 42	FP = 6	31	8	37	9
	FN = 4	TN = 36	9	40	2	40

A. RANDOM FOREST CLASSIFIER

The Random Forest classifier has the highest accuracy and precision for the ECG signal with an accuracy of 88.64% and precision of 87.50%. It has a recall of 91.30%. For the fingertip-PPG signal, it shows an accuracy of 80.68% and

precision and recall of 79.48% and 77.50%, respectively. For the ECG + fingertip-PPG signal, it shows an accuracy of 87.50% with precision and recall of 80.43% and 94.87%. The confusion matrix shows a higher number of true positives for the ECG signal, but higher true negatives for the fingertip-PPG signal and ECG + fingertip-PPG signal.

B. SVM CLASSIFIER

The SVM classifier has the highest accuracy and recall for the ECG signal with an accuracy of 51.14% and recall of 64.70%. It has a precision of 22.91%. For the fingertip-PPG signal, it shows an accuracy of 46.59% and recall of 44.28%, and it has the highest precision of 79.48%. For the ECG + fingertip-PPG signal, it shows an accuracy of 50% with precision and recall of 10.86% and 62.50%, respectively. The confusion matrix shows a higher number of false positives for the ECG signal and ECG + fingertip-PPG signal, but higher false negatives for the fingertip-PPG signal.

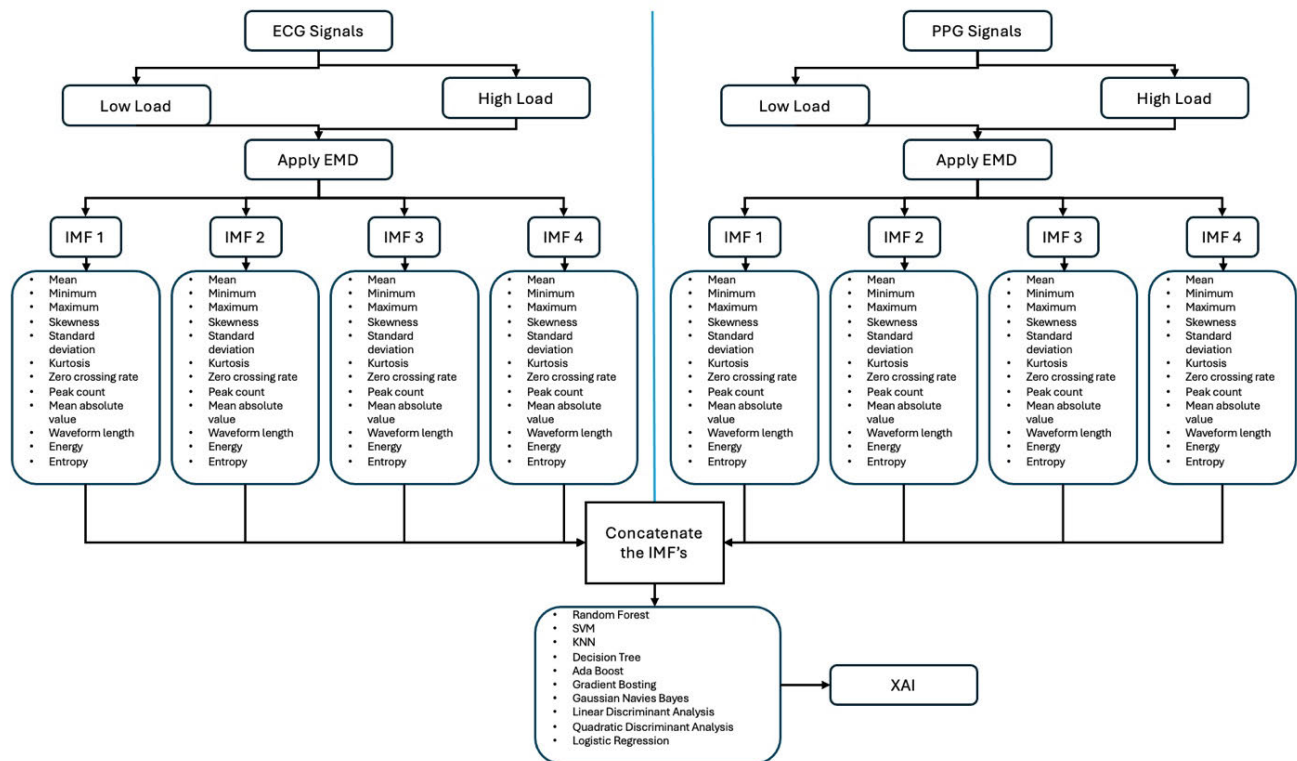


FIGURE 12. Methodology of ECG signal and PPG signal as multimodal data.

C. KNN CLASSIFIER

The KNN classifier shows the highest accuracy, precision, and recall for the ECG signal, with an accuracy of 80.68%, precision of 72.91%, and recall of 89.74%. It shows an accuracy of 64.77%, precision of 69.23%, and recall of 58.69% for the fingertip-PPG signal. For the ECG + fingertip-PPG signal, it has an accuracy of 79.55% with precision and recall of 71.73% and 86.84%, respectively. The confusion matrix shows higher numbers of true negatives.

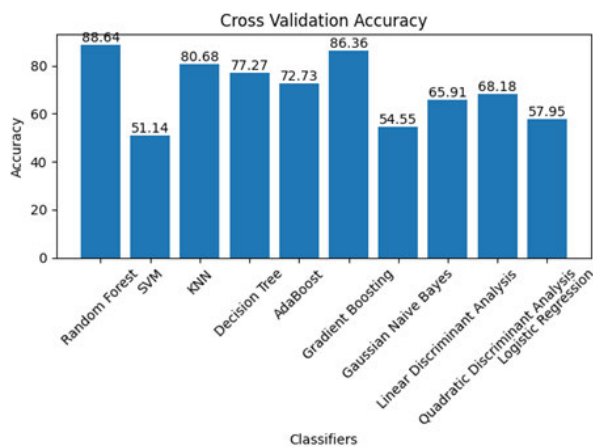


FIGURE 13. Accuracy comparison between classifiers for ECG signal.

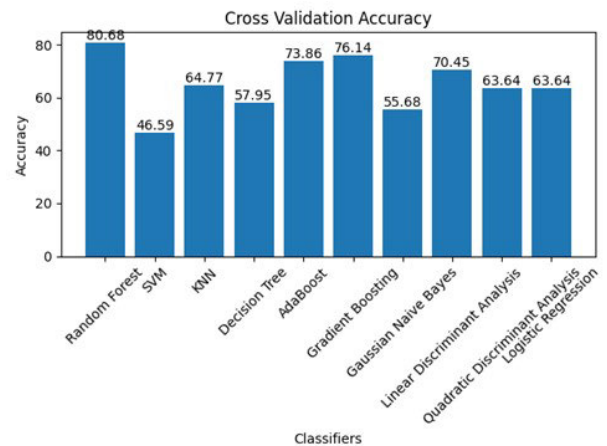


FIGURE 14. Accuracy comparison between classifiers for fingertip-PPG signal.

D. DECISION TREE CLASSIFIER

The Decision Tree classifier shows the highest accuracy, precision, and recall for the ECG signal, with an accuracy of 77.27%, precision of 81.25%, and recall of 78%. It shows an accuracy of 57.95%, precision of 51.28%, and recall of 52.63% for the fingertip-PPG signal. For the ECG + fingertip-PPG signal, it has an accuracy of 68.18% with precision and recall of 67.39% and 70.45%, respectively. The

TABLE 2. Recent works and proposed work on MAUS dataset.

Author	Year	Features	Methods	Modality	Accuracy
Win-Ken Beh, Yi-Hsuan Wu, An-Yeu Wu [16]	2021	HRV (Time Domain and Frequency Domain)	SVM-LOSO	ECG, Fingertip-PPG, Wrist-PPG	71.6% 66.7% 59.9%
			SVM-Mixed Subject 5-Fold	ECG, Fingertip-PPG, Wrist-PPG	71.7% 70.8% 64.8%
Amanrosekaur Sethi, Rahee Walambe, Priyanka Jain, Ketan Kotecha [21]	2023	HRV (Time Domain and Frequency Domain)	Ada-Boost Classifier	ECG + Fingertip-PPG + Wrist-PPG	75.7%
Win-Ken Beh, Yi-Hsuan Wu, An-Yeu Wu [22]	2023	HRV (Time Domain, Frequency Domain and Non-Linear Domain)	Linear-SVM	MW System (PPG)	66.7%
				MW System + Outlier Removal (PPG)	69.4%
				MW System + Outlier Removal + Uncertainty Estimation (PPG)	74.2%
				ECG	75.3%
Proposed Method	2024	EMD Features Extractions	Random Forest	ECG	88.64%
				Fingertip-PPG	80.68%
				ECG + Fingertip-PPG	87.50%

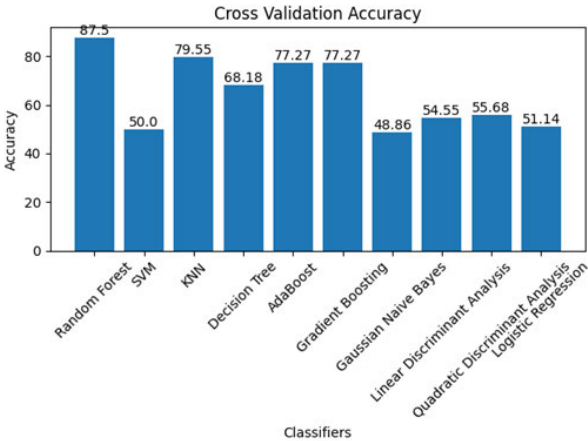


FIGURE 15. Accuracy comparison between classifiers for ECG + fingertip-PPG signal.

confusion matrix shows higher numbers of true positives for the ECG signal and ECG + fingertip-PPG signal and higher true negatives for the fingertip-PPG signal.

E. ADABOOST CLASSIFIER

The AdaBoost classifier shows the highest accuracy and precision for the ECG + fingertip-PPG signal, with an accuracy of 77.27% and precision of 78.26%. It has a recall of 78.26%. It shows the highest recall of 81.57% with an accuracy and precision of 72.73% and 64.58%, respectively, for the ECG signal. For the fingertip-PPG signal, it shows accuracy, precision, and recall of 77.27%, 74.35%, and 69.04%, respectively. The confusion matrix shows a higher number of true negatives for the ECG and fingertip-PPG signals but shows a higher number of true positives for the ECG + fingertip-PPG signal.

F. GRADIENT BOOSTING CLASSIFIER

The Gradient Boosting classifier shows the highest accuracy, precision, and recall of 86.36%, 89.58%, and 86%, respectively, for the ECG signal. It shows accuracy, precision, and recall of 76.14%, 79.48%, and 70.45% for the fingertip-PPG signal, respectively. For the ECG + fingertip-PPG signal, it shows an accuracy of 77.27% with precision and recall of 73.91% and 80.95%. The confusion matrix shows a higher number of true positives and true negatives for the ECG signal and fingertip-PPG signal, respectively, and higher and equal numbers of true positives and true negatives for the ECG + fingertip-PPG signal.

G. GAUSSIAN NAIVE BAYES CLASSIFIER

The Gaussian Naive Bayes classifier shows higher accuracy and precision for the fingertip-PPG signal, with an accuracy of 55.68% and precision of 64.10%. It has a recall of 50%. It shows a higher recall of 61.11% for the ECG signal with an accuracy of 54.55% and precision of 45.83%. For the ECG + fingertip-PPG signal, it shows an accuracy of 48.86% with precision and recall of 32.60% and 51.72%, respectively. The confusion matrix shows higher and equal numbers of false positives and true negatives for the ECG signal and higher and equal numbers of true positives and false negatives for the fingertip-PPG signal, and it has a higher number of false positives for the ECG + fingertip-PPG signal.

H. LINEAR DISCRIMINANT ANALYSIS CLASSIFIER

The Linear Discriminant Analysis classifier shows higher accuracy and precision for the fingertip-PPG signal, with an accuracy of 70.45% and precision of 71.79%. It has a recall of 65.11%. It shows a higher recall of 71.42% for the ECG signal with an accuracy of 65.91% and precision of 62.50%. For the ECG + fingertip-PPG signal, it shows an accuracy

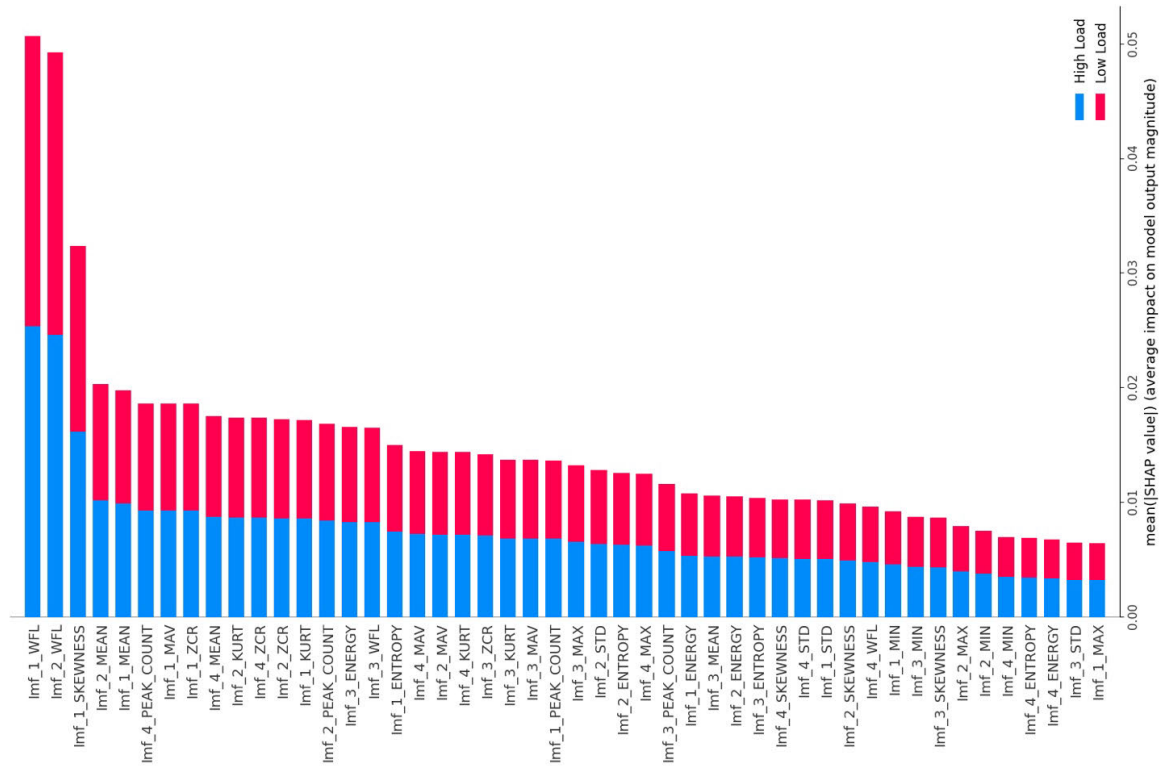


FIGURE 16. SHAP analysis results for ECG signal.

of 54.55% with precision and recall of 56.52% and 56.52%, respectively. The confusion matrix shows a higher number of true positives for the ECG signal and ECG + fingertip-PPG signal and a higher number of true negatives for the fingertip-PPG signal.

I. QUADRATIC DISCRIMINANT ANALYSIS CLASSIFIER

The Quadratic Discriminant Analysis classifier shows the highest accuracy and recall of 68.18% and 76.31% for the ECG signal. It has a precision of 60.41%. It shows accuracy, precision, and recall of 63.64%, 69.23%, and 57.44% for the fingertip-PPG signal, respectively. For the ECG + fingertip-PPG signals, it shows the highest precision of 71.73% with an accuracy and recall of 55.68% and 55.93%, respectively. The confusion matrix shows a higher number of true negatives for the ECG signal and fingertip-PPG signal, respectively, and a higher number of true positives for the ECG + fingertip-PPG signal.

J. LOGISTIC REGRESSION CLASSIFIER

The Logistic Regression classifier shows higher accuracy and precision for the fingertip-PPG signal, with an accuracy of 63.64% and precision of 64.10%. It has a recall of 58.13%. It shows a higher recall of 63.41% for the ECG signal with an accuracy of 57.95% and precision of 54.16%. For the ECG + fingertip-PPG signal, it shows an accuracy of 51.14% with precision and recall of 43.47% and 54.05%, respectively. The

confusion matrix shows a higher number of true positives for the ECG signal, true negatives for the fingertip-PPG signal, and false positives for the ECG + fingertip-PPG signal.

For comparison with our proposed method, we also applied 1D-CNN with MAUS dataset, as they are known for their high accuracy in stress detection, however the 1D-CNN haven't performed better as compared to the Machine Learning classifiers on this dataset.

Fig.13-15 shows the accuracy comparison between classifiers for ECG signal, PPG signal and ECG + PPG Signals. Table 1 shows the performance metrics for random forest classifier and Table 2 shows the recent works and proposed work on MAUS dataset.

VIII. SHAP ANALYSIS RESULTS

The chart shows each feature's impact under high and low mental load conditions, with some features showing a greater effect under one condition than the other. Each bar represents the mean absolute SHAP value, reflecting the average contribution of a feature to the model's predictions. The features are derived from the first four Intrinsic Mode Functions (IMFs) obtained via Empirical Mode Decomposition (EMD) of the ECG signal, fingertip-PPG Signal, and ECG + fingertip-PPG Signal. Features are labeled with their corresponding IMF and the specific metric used (e.g., WFL for waveform length, SKEWNESS, MEAN, ZCR for zero-crossing rate, etc.). The bars are color-coded, where blue indicates the feature's

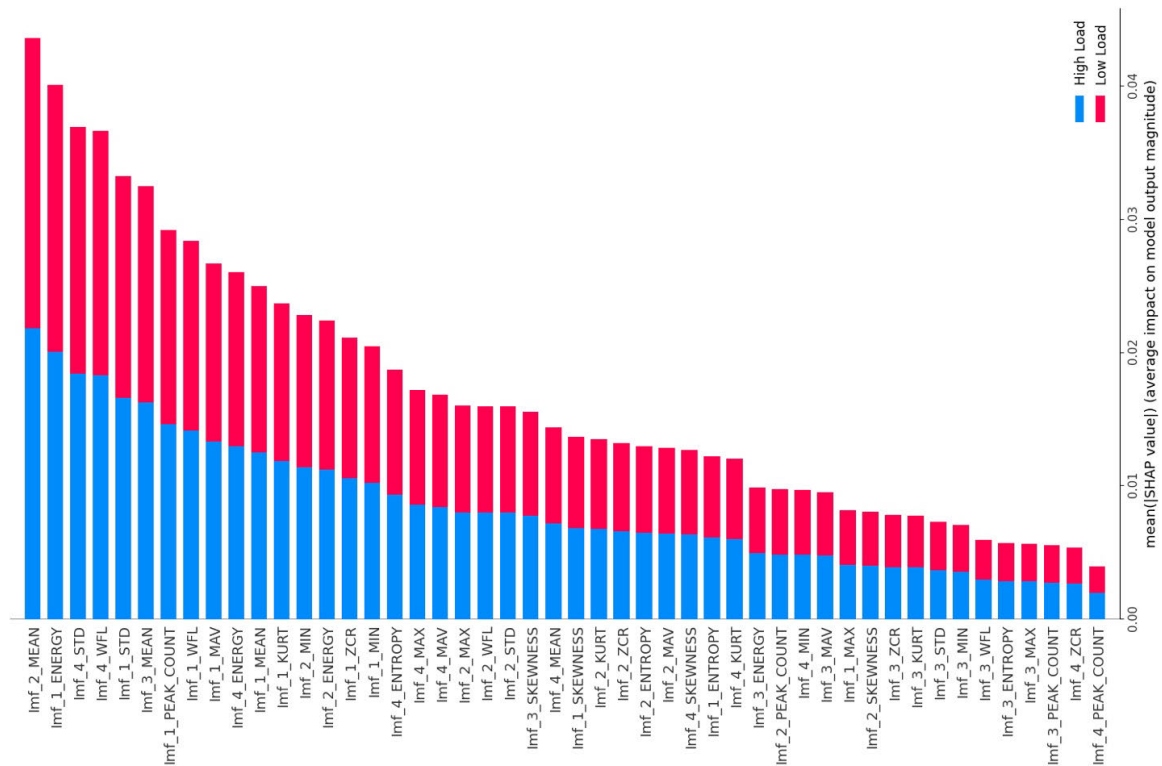


FIGURE 17. SHAP analysis results for fingertip-PPG signal.

impact under High Load conditions while red indicates the feature's impact under Low Load conditions.

In addition to evaluating performance metrics, SHAP analysis provided insights into the individual contributions of features to the model's predictions. For ECG signals, *Imf_1_WFL* (waveform length of the first IMF) was the most critical feature for distinguishing between low and high mental workload, capturing variations in cardiac activity that are indicative of stress levels. Similarly, for fingertip-PPG signals, the mean value of the second IMF (*Imf_2_MEAN*) was the most significant, indicating changes in peripheral circulation associated with stress. When combining ECG and fingertip-PPG as multimodal data, *Imf_2_WFL* (the waveform length of the second IMF) had the highest SHAP value, highlighting the multimodal approach's benefit for enhancing stress detection capabilities. This combined model improved recall and provided deeper insights into the contributions of different physiological signals to mental workload classification.

A. ECG SIGNAL

Features like waveform length, skewness, and mean of the IMFs have significant contributions to the model's ability to distinguish between high and low mental workload. The most influential features, as indicated by the longer bars, include '*Imf_1_WFL*' (Waveform length of the first IMF), '*Imf_2_WFL*' (Waveform length of the second

IMF), '*Imf_1_SKEWNESS*' (Skewness of the first IMF), '*Imf_2_MEAN*' (Mean value of the second IMF), and '*Imf_1_MEAN*' (Mean value of the first IMF).

B. FINGERTIP-PPG SIGNAL

Features like mean value, energy, and standard deviation of the IMFs have significant contributions to the model's ability to distinguish between high and low mental workload. The most influential features, as shown by the longer bars, include '*Imf_2_MEAN*' (Mean value of the second IMF), '*Imf_1_ENERGY*' (Energy of the first IMF), '*Imf_4_STD*' (Standard deviation of the fourth IMF), '*Imf_4_WFL*' (Waveform length of the fourth IMF), and '*Imf_1_STD*' (Standard deviation of the first IMF).

C. COMBINED ECG + FINGERTIP-PPG SIGNAL

Features like waveform length, zero-crossing rate, peak count, and skewness of the IMFs have significant contributions to the model's ability to distinguish between high and low mental workload. The most influential features, as represented by the longer bars, include '*Imf_2_WFL*' (Waveform length of the second IMF), '*Imf_1_WFL*' (Waveform length of the first IMF), '*Imf_2_ZCR*' (Zero-crossing rate of the second IMF), and '*Imf_4_PEAK_COUNT*' (Peak count of the fourth IMF).

In summary, SHAP analysis charts highlight the most influential features derived from EMD of ECG Signal,

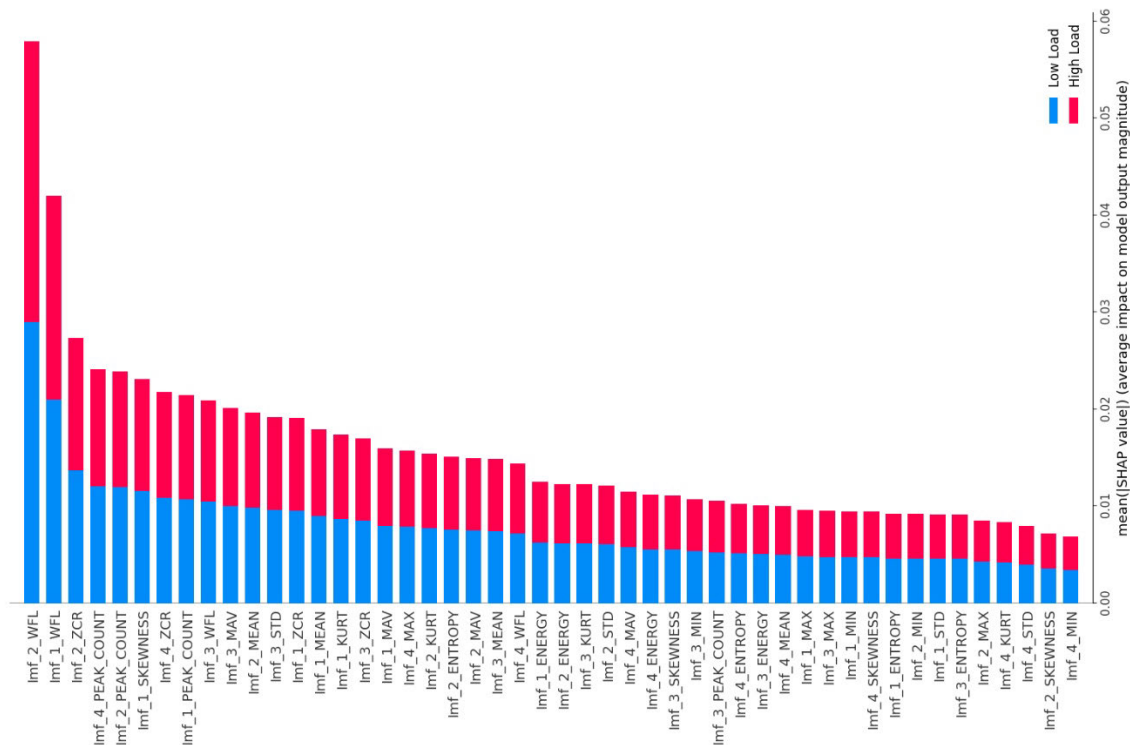


FIGURE 18. SHAP analysis results for ECG + fingertip-PPG signal.

fingertip-PPG Signal, and ECG + fingertip-PPG signals for classifying mental workload, showing their relative importance and differences in impact between high and low load conditions. Fig. 16-18 shows the SHAP analysis results for ECG signal, PPG signal and ECG + PPG signal.

IX. PERFORMANCE EVALUATION AND INSIGHTS

The Random Forest model's performance across different modalities in N-back tasks shows differing levels of accuracy and effectiveness. For ECG recordings, the model achieved an accuracy of 88.64%, with a precision of 0.875 and a recall of 0.913. The confusion matrix reveals that the model correctly identified 42 true positives and 36 true negatives while misclassifying 6 false positives and 4 false negatives. In contrast, the fingertip-PPG model showed a lower accuracy of 80.68%, with a precision of 0.795 and a recall of 0.775, yielding 31 true positives and 40 true negatives, alongside 8 false positives and 9 false negatives. When combining ECG and fingertip-PPG data, the model's accuracy increased to 87.5%, with a precision of 0.804 and a recall of 0.949. The confusion matrix for this multimodal approach yielded 37 true positives and 40 true negatives, with 9 false positives and 2 false negatives.

This result shows that combining both modalities (ECG signal and fingertip-PPG signal) can provide complementary insights for improving the model's ability to detect true positives, particularly in scenarios involving high cognitive

load. The confusion matrices for each modality indicated that the model performed well in detecting true positives, but there were some false positives, especially in the multimodal approach. The SHAP analysis highlighted this by showing that features like waveform length and mean value play a dominant role."

SHAP analysis further illuminated the feature contributions: for ECG, the waveform length of the first IMF (Imf_1_WFL) was the most influential, whereas for fingertip-PPG, the mean of the second IMF (Imf_2_MEAN) was most impactful. In the combined ECG + fingertip-PPG model, the waveform length of the second IMF (Imf_2_WFL) emerged as the most significant feature. Although the combined model showed slightly lower accuracy compared to ECG alone, it improved recall, demonstrating better performance in identifying true positives. Overall, while accuracy gives a general measure of model performance, the confusion matrix and SHAP analysis offer deeper insights into the model's strengths and areas for improvement, emphasizing the value of multimodal data in improving classification outcomes.

X. CONCLUSION

This study presents a method for mental workload detection by utilizing ECG and fingertip-PPG signals, with Empirical Mode Decomposition (EMD) used to extract insightful features from these physiological signals. The

approach demonstrates the effectiveness of EMD-based features extracted from ECG and fingertip-PPG recordings across various machine learning classifiers. The Random Forest classifier achieved notable performance, with an accuracy of 88.64% for ECG alone and 80.68% for fingertip-PPG. When combining ECG and fingertip-PPG data, it led to an accuracy of 87.5% and a notable improvement in recall, reaching 94.9%. This study's use of SHAP analysis highlighted the importance of features such as waveform length and mean values of the IMFs, which are essential in distinguishing between high and low mental workload conditions. The critical features that significantly influence workload classification are: the waveform length of the first IMF (Imf_1_WFL) for ECG, the mean of the second IMF (Imf_2_MEAN) for fingertip-PPG, and the waveform length of the second IMF (Imf_2_WFL) for ECG + fingertip-PPG Signal. While the study employed traditional machine learning methods rather than deep learning, the results highlight the potential of EMD-derived features for enhancing classification performance. Future research will focus on integrating additional physiological signals and larger datasets to further improve accuracy and robustness in mental workload detection.

ACKNOWLEDGMENT

(Aakanksha Dayal and Ziaullah Khan contributed equally to this work.)

DECLARATIONS

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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